Unit 2 - Linear Algebra
2.2 Vectors

Graham West

MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Why Vectors?
2. Vectors as Arrows, Rows, and Columns
3. Vector Operations
4. The Dot Product and the Angle
5. Linear Combinations, Span, and Independence
6. Basis and Change of Basis
7. Looking Ahead: Where Matrices Come From
8. Review

Why Vectors?

Where This Shows Up in Data Science

A row of a data table is a vector

- **Every record is a point.** One patient, one house, one image, one document becomes a list of numbers, and the whole dataset becomes a cloud of points.
- **Similarity is geometry.** Recommenders, search, and duplicate detection all rank things by **cosine similarity**, which is an angle.
- **Distance is the basis of clustering.** k-means, nearest neighbors, and anomaly detection all measure how far apart two vectors are.
- **Redundant features are dependent vectors.** When two columns carry the same information, their vectors lie on one line. That is **collinearity**, and it wrecks a regression.
- **Compression is choosing a better basis.** PCA, at the end of this unit, is exactly that.

Unit 2-1 wrote systems as grids of numbers. Here we ask what a **column** of numbers means on its own.

What We Will Build

The plan

1. A **vector**, drawn as an arrow and written as a column (or row) of numbers.
2. A vector's **magnitude**; **adding**, **scaling**, and **subtracting** vectors; the **distance** between points; and **unit vectors**.
3. The **dot product**: one number that measures how much two vectors agree. From it: the **angle**, **cosine similarity** (how a spam filter tells which emails are alike), **orthogonality**, and **projection**.
4. **Linear combinations**, **span**, and **independence**: which points you can reach, and whether anything is redundant.
5. **Basis**, **dimension**, **change of basis**, and a look ahead to **matrices**.

Notebooks: `U2-2_Vectors-1_ArrowsAndOps`, `-2_DotProductAndAngle`, `-3_ProjectionAndSimilarity`, `-4_LinearCombinationsAndSpan`, `-5_IndependenceAndBasis`.

Vectors as Arrows, Rows, and Columns

Two Pictures of the Same Thing

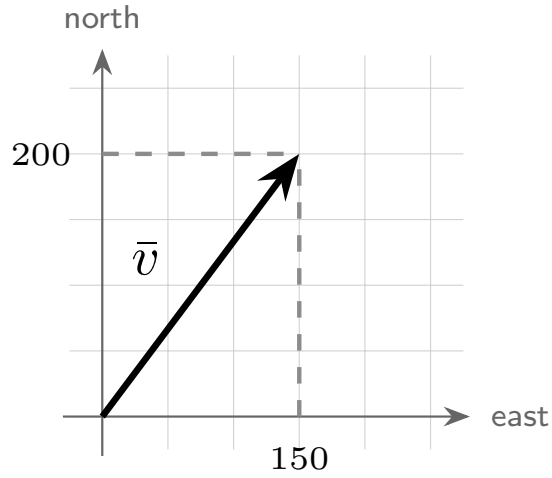
Definition

A **vector** is a quantity with both a **magnitude** (how big) and a **direction** (which way).

Size only	*Size and direction: a vector*
distance: 5 km	**displacement**: 5 km north
speed: 250 m/s	**velocity**: 150 east + 200 north (m/s)

A plane's velocity is **drawn** as an arrow and **written** as components, one per axis:

$$\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix} \begin{matrix} \leftarrow \text{east} \\ \leftarrow \text{north} \end{matrix}$$



Velocity in m/s; one square is 50 m/s.

Notation: marking a vector

- $\mathbf{v}$, $\bar{v}$, $\vec{v}$ all mean "vector": bold when typing, bar or arrow by hand (we use the **bar**).
- $\hat{v}$ (hat) means a vector of length exactly 1, a **unit vector**.

Notebook `U2-2_Vectors-1_ArrowsAndOps`, §1, the arrow and the column side by side

Three Running Examples

Three vectors will follow us through the whole lecture:

Velocity

A plane flying 150 m/s east and 200 m/s north:

$$\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$$

One number per direction.

Patient

Age, systolic blood pressure, A1c:

$$\bar{p} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix}$$

One number per measurement.

Email

Counts of *free*, *win*, *cash*, *meeting*:

$$\bar{d}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

One number per word.

The question they build toward

- Two patients, or two emails: **how similar are they?** Is a new email **spam**?
- By the end, "similar" is a number we can **compute**: how **close** two vectors are, or how nearly they **point the same way**.

Turning an Email into a Vector

A **spam filter** compares emails, so first it turns each one into numbers.

Algorithm: a count vector

1. Fix a **vocabulary**: the words to track.
2. In each email, **count** each word.
3. The list of counts **is** the vector.

- Order and capitals are thrown away: *Free* counts as *free*.
- Real filters track thousands of words: $\mathbb{R}^{5000}$ or more.

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
free	2	4	2
win	1	2	0
cash	2	4	1
meeting	0	0	2

Each **column** is one email's vector.

Three emails

$\bar{d}_1$	"Win cash now! Free cash, free entry."	**spam**
$\bar{d}_2$	"Win cash now! Free cash, free entry. Win free cash, free prizes, cash today."	**spam**
$\bar{d}_3$	"Free lunch at the meeting. After the meeting, free parking; bring cash."	**not spam**

Columns and Rows: Two Ways to Write a Vector

Column or row: same numbers, different shape

$$\bar{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \text{ (**column**, } 2 \times 1) \qquad \bar{v} = \begin{bmatrix} 3 & 4 \end{bmatrix} \text{ (**row**, } 1 \times 2)$$

- Same information, **different shapes**: $n \times 1$ or $1 \times n$. From Unit 2-3 on, the shape decides what can be multiplied by what.
- We write vectors as **columns by default**; a patient stored in a data table is a **row**.
- A **point** is neither: it stays a tuple, $(3, 4)$.

In numpy: three kinds of array

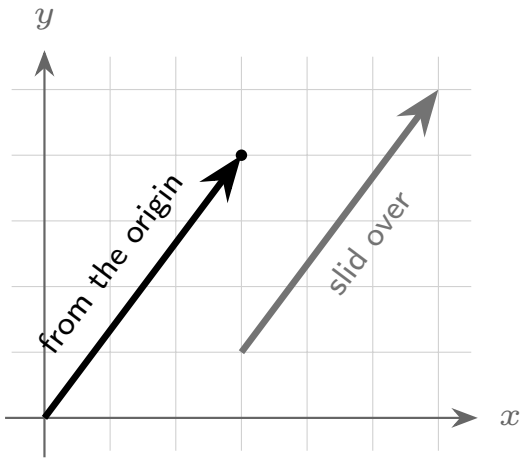
Shape	**Built with**	**What it is**
`(n,)`	`np.array([3, 4])`	a flat list, **no orientation**
`(n,1)`	`np.array([[3], [4]])`	a **column**: n rows, 1 entry each
`(1,n)`	`np.array([[3, 4]])`	a **row**: 1 row, n entries

Check with `v.shape`. Most code uses the flat `(n,)` kind.

Point Versus Vector

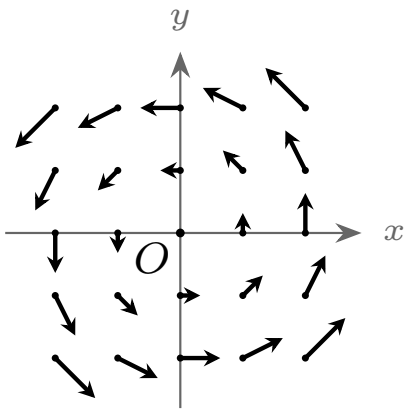
Same numbers, different ideas

- A **point** $(3, 4)$ is a **location**. It sits where it sits.
- A **vector** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is a **displacement**: 3 across, 4 up. It has **no home**.



Two drawings of the **same** vector. Only the dot is a point.

- We draw vectors from the origin **by default**, so arrow and point look alike.
- Sliding an arrow, same length and direction, **changes nothing**.
- Addition, next section, relies on that freedom.



A **velocity field**: fluid swirling around the origin O. Each arrow sits **where the fluid is**, and the fluid moves faster farther out.

Beyond the Plane

Nothing about the definition needs two components. A vector with n components lives in $\mathbb{R}^n$, the set of all lists of n real numbers.

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

The running examples, in any dimension

- **Velocity**: add "up" for a plane that is climbing, and it lives in $\mathbb{R}^3$.
- **Patient**: 3 measurements is $\mathbb{R}^3$; a 40-value lab panel is $\mathbb{R}^{40}$.
- **Email**: a 5000-word vocabulary puts every email in $\mathbb{R}^{5000}$.

The habit to build

- You cannot **picture** $\mathbb{R}^{40}$, and you do not need to.
- Every formula here is a sum over components, so it works for **any** n.
- Reason with the picture in $\mathbb{R}^2$; trust the algebra in $\mathbb{R}^n$.

Vector Operations

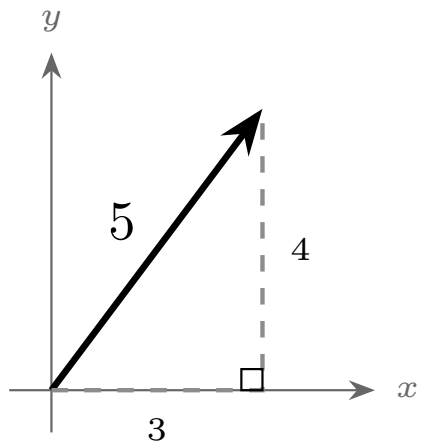
Magnitude: How Long Is a Vector?

Definition

The **magnitude**, or **norm**, of $\bar{v} \in \mathbb{R}^n$:

$$\|\bar{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Never negative; zero only for $\bar{0}$.



Legs 3 and 4, hypotenuse 5.

Running examples

- **Velocity**: $\|\bar{v}\| = \sqrt{150^2 + 200^2} = 250$ m/s, the plane's **speed**.
- **Email**: $\|\bar{d}_1\| = \sqrt{4 + 1 + 4 + 0} = 3$ and $\|\bar{d}_2\| = 6$: longer email, longer vector.
- **Patient**: $\|\bar{p}\|$ mixes years, mmHg, and percent, so it is **meaningless**.

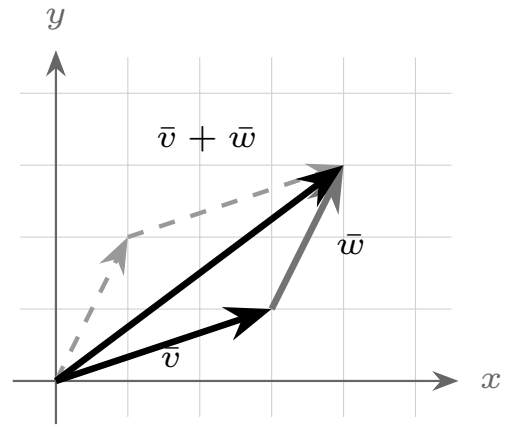
Notebook `U2-2_Vectors-1_ArrowsAndOps`, §5, magnitude as a hypotenuse

Addition: Head to Tail

Componentwise, as you would guess:

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

- Tail of $\bar{w}$ on the head of $\bar{v}$.
- The sum: first tail to last head.
- Either order: $\bar{v} + \bar{w} = \bar{w} + \bar{v}$.



Either route reaches $(4, 3)$.

Running examples

- **Velocity**: a plane's velocity through the air $\bar{v}$, plus the wind $\bar{w}$, is its velocity over the **ground**, $\bar{v} + \bar{w}$.
- **Email**: paste $\bar{d}_1$ and $\bar{d}_3$ together and the counts add: 4 *free*, 1 *win*, 3 *cash*, 2 *meeting*.

The triangle inequality

$$\|\bar{v} + \bar{w}\| \leq \|\bar{v}\| + \|\bar{w}\|$$

Here $5 \leq \sqrt{10} + \sqrt{5} \approx 5.4$: the direct route wins.

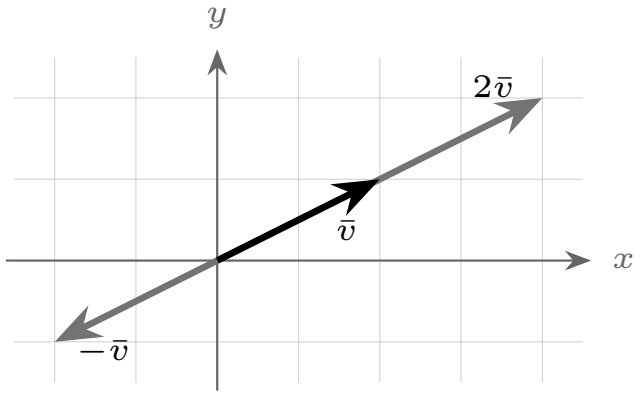
Notebook `U2-2_Vectors-1_ArrowsAndOps`, §2, addition drawn head to tail

Scalar Multiplication: Stretch, Shrink, Flip

A **scalar** is a plain number. Multiplying by it **scales**:

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c\,v_1 \\ c\,v_2 \end{bmatrix}$$

- $c > 1$ stretches; $0 < c < 1$ shrinks.
- $c < 0$ flips; $c = 0$ gives $\bar{0}$.
- Every multiple stays on **one line through the origin**: remember this for **span**.



All multiples of $\bar{v}$ lie on one line.

Running examples

- **Velocity**: $2\bar{v}$ doubles the speed on the same heading; $-\bar{v}$ reverses course.
- **Units**: m/s to km/h is scaling by 3.6, so 250 m/s is 900 km/h.
- **Email**: $\bar{d}_2 = 2\bar{d}_1$: every word twice as often. Same **direction**, twice the length.

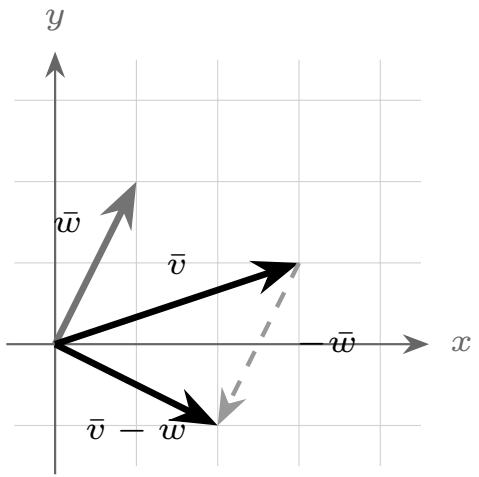
Subtraction: Add the Reverse

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Addition in disguise

$\bar{v} - \bar{w} = \bar{v} + (-\bar{w})$

- $-\bar{w}$: same length, opposite way.
- Put its tail on the head of $\bar{v}$.



$\bar{v}$, then the reversed $\bar{w}$ (dashed).

Velocity: a plane in a headwind

A plane flies 250 m/s east; a 50 m/s **headwind** takes some away:

$$\begin{bmatrix} 250 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 0 \end{bmatrix} = \begin{bmatrix} 200 \\ 0 \end{bmatrix}$$

Its velocity over the **ground**.

Patient: what changed

A year later, minus today:

$$\begin{bmatrix} 55 \\ 136 \\ 6.5 \end{bmatrix} - \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 0.4 \end{bmatrix}$$

$\bar{p}' - \bar{p}$: up 1 year, 6 mmHg, 0.4 A1c.

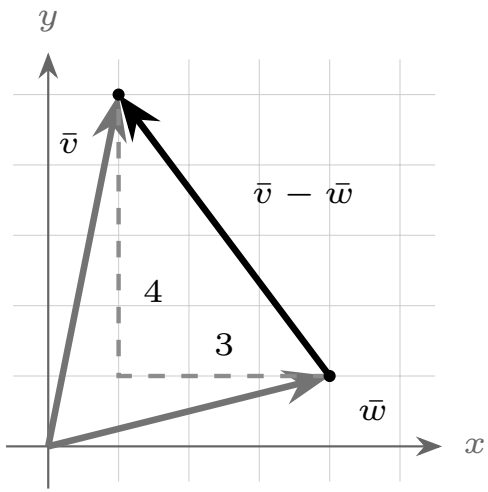
Distance Between Two Points

Definition

The **distance** between two points is the length of the trip between them:

$$d(\bar{v}, \bar{w}) = \|\bar{v} - \bar{w}\| = \sqrt{\sum_i (v_i - w_i)^2}$$

Euclidean distance: what k-means and nearest-neighbor methods measure.



Legs 3 and 4: the points are 5 apart.

Running example: $\bar{p}$ and a second patient $\bar{q}$, and a warning

$$\bar{p} - \bar{q} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} - \begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix} = \begin{bmatrix} -4 \\ -8 \\ -0.3 \end{bmatrix}, \qquad d(\bar{p}, \bar{q}) = \sqrt{16 + 64 + 0.09} \approx 8.95$$

- Blood pressure supplies 64 of the 80; A1c supplies 0.09 and barely registers.
- The biggest units win, so **standardize** before measuring distance.

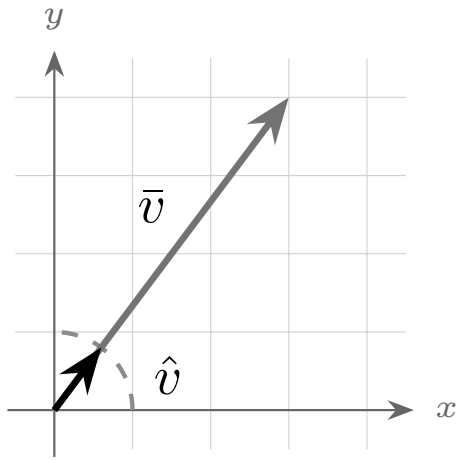
Unit Vectors: Getting $\hat{v}$ from $\bar{v}$

A **unit vector** has length exactly 1; the **hat** marks one. $\hat{v}$ keeps the **direction** of $\bar{v}$ and drops its size.

Normalizing: divide by the length

$$\hat{v} = \frac{1}{\|\bar{v}\|}\,\bar{v} = \frac{\bar{v}}{\|\bar{v}\|}$$

- **Scaling** by $1/\|\bar{v}\|$: same direction, length 1.
- Not defined for $\bar{0}$, which has no direction.



$\hat{v}$ ends on the arc of radius 1.

Running examples

- **Velocity**: the **heading** is $\hat{v} = \frac{1}{250}\begin{bmatrix}150\\200\end{bmatrix} = \begin{bmatrix}0.6\\0.8\end{bmatrix}$; check $0.6^2 + 0.8^2 = 1$.
- **Email**: $\bar{d}_2 = 2\bar{d}_1$, so $\hat{d}_2 = \hat{d}_1$: same mix of words, length forgotten.

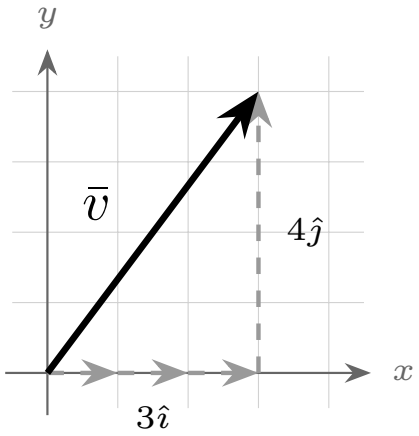
The Standard Unit Vectors

One step along each axis:

$$\hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Scale and add them to build **every** vector:

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3\hat{\imath} + 4\hat{\jmath}$$



The components are the **coefficients** of $\hat{\imath}$ and $\hat{\jmath}$: a **choice of axes**, which we will later change.

Running examples

- **Velocity**: $150\hat{\imath} + 200\hat{\jmath}$ reads "150 m/s east plus 200 m/s north."
- **Email**: each word gets its own unit vector, an email saying just that word once. The counts are the coefficients.

Notebook `U2-2_Vectors-1_ArrowsAndOps`, §4, building $\bar{v}$ from $\hat{\imath}$ and $\hat{\jmath}$

Exercise: Vector Arithmetic

Try it

Let $\bar{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\bar{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

1. Compute $\bar{u} + \bar{v}$ and $\bar{u} - \bar{v}$.
2. Compute $2\bar{u} - 3\bar{v}$.
3. Compute $\|\bar{u}\|$.
4. Find the distance between the points at the heads of $\bar{u}$ and $\bar{v}$.
5. Write $\bar{u}$ in terms of $\hat{\imath}$ and $\hat{\jmath}$.
6. What goes wrong if you try to add $\bar{u}$ to $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$?

Exercise: Solution

Answers

1. $\bar{u} + \bar{v} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $\bar{u} - \bar{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Add and subtract component by component.
2. $2\bar{u} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $3\bar{v} = \begin{bmatrix} 12 \\ -3 \end{bmatrix}$, so $2\bar{u} - 3\bar{v} = \begin{bmatrix} -10 \\ 9 \end{bmatrix}$. Scale first, then subtract.
3. $\|\bar{u}\| = \sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$. Leave it exact unless a decimal is asked for.
4. From part 1, $\bar{u} - \bar{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, so the distance is $\sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5$. A 3-4-5 triangle again.
5. $\bar{u} = 1\hat{\imath} + 3\hat{\jmath}$.
6. **Nothing is defined.** $\bar{u}$ lives in $\mathbb{R}^2$, the other in $\mathbb{R}^3$: no third component of $\bar{u}$ to add to the 3. Operations need **matching dimensions**; in `numpy` this raises a broadcast error.

The Dot Product and the Angle

One Number That Compares Two Vectors

Definition

For $\bar{v}, \bar{w} \in \mathbb{R}^n$, the **dot product** (or **inner product**) is

$$\bar{v} \cdot \bar{w} \;=\; v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \;=\; \sum_{i=1}^{n} v_i w_i$$

How to read it

- Multiply matching components, then add.
- The result is a **number**, not a vector.
- Both need the **same** number of components.

Running example: two emails

$$\bar{d}_1 \cdot \bar{d}_3 = (2)(2) + (1)(0) + (2)(1) + (0)(2) = 6$$

- Each word adds (count here) $\times$ (count there).
- Emails with **no words in common** score 0.

Notebook `U2-2_Vectors-2_DotProductAndAngle`, §1, the dot product by hand and with `np.dot`

Careful: Which Multiplication?

"Multiply two vectors" is ambiguous: the **symbol** decides

Written	**Name**	**Gives**
$c\,\bar{v}$	scalar multiplication	a **vector** (each entry times c)
$\bar{v} \cdot \bar{w}$	dot product	a **number**
$\bar{v} \times \bar{w}$	cross product	a **vector**, only in $\mathbb{R}^3$ (not in this course)
$\bar{v}\,\bar{w}$	side by side	**undefined** for two column vectors

Be careful

- Side by side, $\bar{v}\,\bar{w}$, is **not** the dot product. Write the dot.
- Side by side gets a meaning in Unit 2-3, for matrices.

In code, with the emails

- `d1 @ d3` is the dot product: `6`.
- `d1 * d3` is **not**: it multiplies entry by entry, `[4 0 2 0]`.

Rules, and What the Sign Tells You

Properties

- $\bar{v} \cdot \bar{w} = \bar{w} \cdot \bar{v}$
- $\bar{u} \cdot (\bar{v} + \bar{w}) = \bar{u} \cdot \bar{v} + \bar{u} \cdot \bar{w}$
- $(c\bar{v}) \cdot \bar{w} = c(\bar{v} \cdot \bar{w})$
- $\bar{v} \cdot \bar{v} = \|\bar{v}\|^2 \geq 0$

The last rule is the link to **length**:

$$\|\bar{v}\| = \sqrt{\bar{v} \cdot \bar{v}}$$

The sign is a verdict on direction

- **Positive:** they broadly **agree**, within a quarter turn.
- **Zero: perpendicular**. Neither has any component along the other.
- **Negative:** they broadly **oppose**, more than a quarter turn apart.

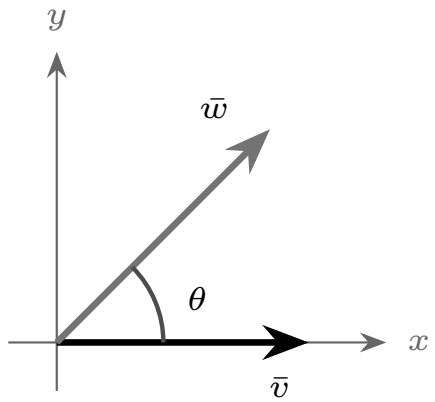
Running example: a plane flying along $\bar{v}$, in a wind $\bar{w}$

- $\bar{v} \cdot \bar{w} > 0$: part **tailwind**. $\bar{v} \cdot \bar{w} < 0$: part **headwind**.
- $\bar{v} \cdot \bar{w} = 0$: pure **crosswind**. No gain or loss along the heading, but it still pushes the plane **sideways**.

The Angle Between Two Vectors

The key formula

$$\cos\theta \;=\; \frac{\bar{v}\cdot\bar{w}}{\|\bar{v}\|\,\|\bar{w}\|}$$



- Dividing by both lengths leaves **direction alone**.
- The result is always between -1 and 1.

Worked: a plane heading east, wind blowing northeast

$$\bar{v}=\begin{bmatrix}1\\0\end{bmatrix},\ \bar{w}=\begin{bmatrix}1\\1\end{bmatrix}: \qquad \cos\theta=\frac{1}{1\cdot\sqrt{2}}=\frac{1}{\sqrt{2}}\ \Rightarrow\ \theta=45^\circ$$

Running examples, beyond the picture

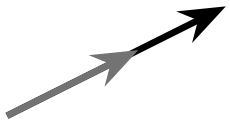
- **Email**: for $\bar{d}_1, \bar{d}_3$, $\cos\theta = 6/(3\cdot 3) = 2/3$, so $\theta \approx 48^\circ$.
- **Patient**: no picture in $\mathbb{R}^{40}$, but the formula still returns a number in $[-1, 1]$.

Cosine Similarity

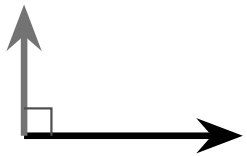
Definition

The **cosine similarity** of nonzero $\bar{v}, \bar{w}$ is the cosine of the angle between them:

$$\cos\theta = \frac{\bar{v} \cdot \bar{w}}{\|\bar{v}\| \, \|\bar{w}\|}$$



$\cos\theta = +1$
same direction



$\cos\theta = 0$
perpendicular, nothing shared



$\cos\theta = -1$
opposite directions

Why it gets its own name

- It sees **direction only**: scaling either vector leaves it unchanged.
- That is exactly right when size is noise, like the **length** of an email.

Notebook `U2-2_Vectors-3_ProjectionAndSimilarity`, §3, a patient similarity heatmap

Why Spam Filters Compare by Cosine

The three emails

$\bar{d}_1$	"Win cash now! Free cash, free entry."	**spam**
$\bar{d}_2$	"Win cash now! Free cash, free entry. Win free cash, free prizes, cash today."	**spam**
$\bar{d}_3$	"Free lunch at the meeting. After the meeting, free parking; bring cash."	**not spam**

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
free	2	4	2
win	1	2	0
cash	2	4	1
meeting	0	0	2

$\|\bar{d}_1\| = \|\bar{d}_3\| = 3, \quad \|\bar{d}_2\| = 6$

Which email is most like the spam $\bar{d}_1$?

- **Distance** picks $\bar{d}_3$, not spam: fooled by **length**.
- **Cosine** picks $\bar{d}_2$: the same spam, longer.

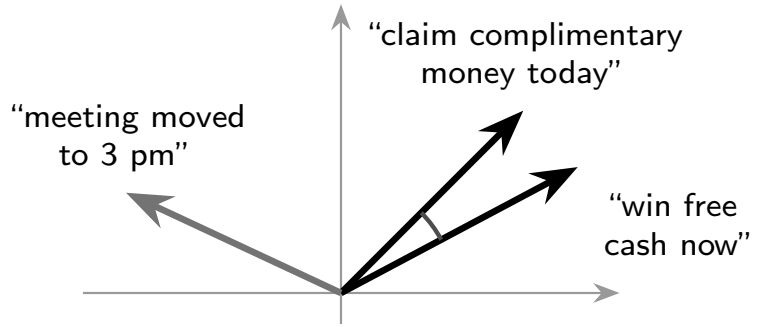
The numbers

	Distance from $\bar{d}_1$	**Cosine with $\bar{d}_1$**
$\bar{d}_2$ (spam, longer)	$\sqrt{2^2 + 1^2 + 2^2 + 0^2} = 3$	$(8 + 2 + 8 + 0)/(3 \cdot 6) = 1$
$\bar{d}_3$ (not spam)	$\sqrt{0^2 + 1^2 + 1^2 + 2^2} \approx 2.45$	$(4 + 0 + 2 + 0)/(3 \cdot 3) \approx 0.67$

Beyond Counting Words: Embeddings

The problem with counting words

- "Win free cash now" and "Claim complimentary money today" mean the **same thing**.
- No shared words, so the count vectors are perpendicular: cosine 0.
- Counts see **spelling**, not **meaning**.



The goal: same meaning, **small angle**.

What we want instead

- "cat" and "dog" should come out **similar**; "cat" and "agglomerative" should not.
- Counts cannot do it: each word is its own unit vector, so **any** two different words are perpendicular.
- We need vectors built from **meaning**. Those are **embeddings**.

What an Embedding Looks Like

The idea: an embedding

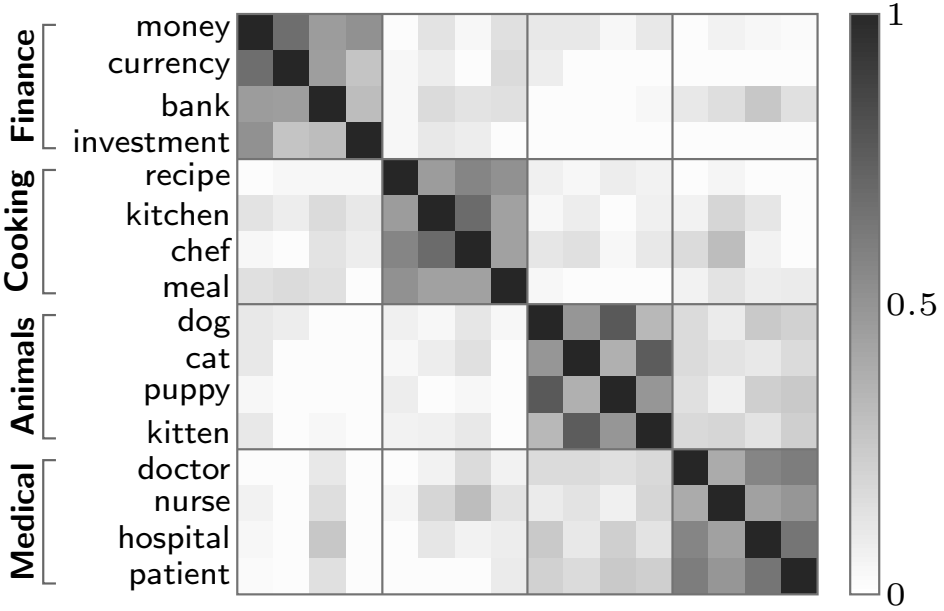
- Each word becomes a vector of **learned** numbers, not counts: hundreds to thousands of components.
- A model learns them by **predicting each word from the words around it**.
- Trained so that **similar meanings point the same way**.

A sentence, one column per word: "The cat sat on the mat"

	the	*cat*	*sat*	*on*	*the*	*mat*
entry 1	0.23	0.32	0.23	−2.63	0.23	0.01
2	0.51	−1.47	0.00	−0.77	0.51	1.99
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$n-1$	−0.21	0.86	−1.47	0.23	−0.21	0.23
n	1.99	0.10	0.51	0.11	1.99	0.50

- Each column is one word's **vector**.
- Both copies of *the* get the **same** column: the vector belongs to the word.
- Side by side, the columns form a grid of numbers, a **matrix**: Unit 2-3.

Cosine Similarity Between Word Embeddings



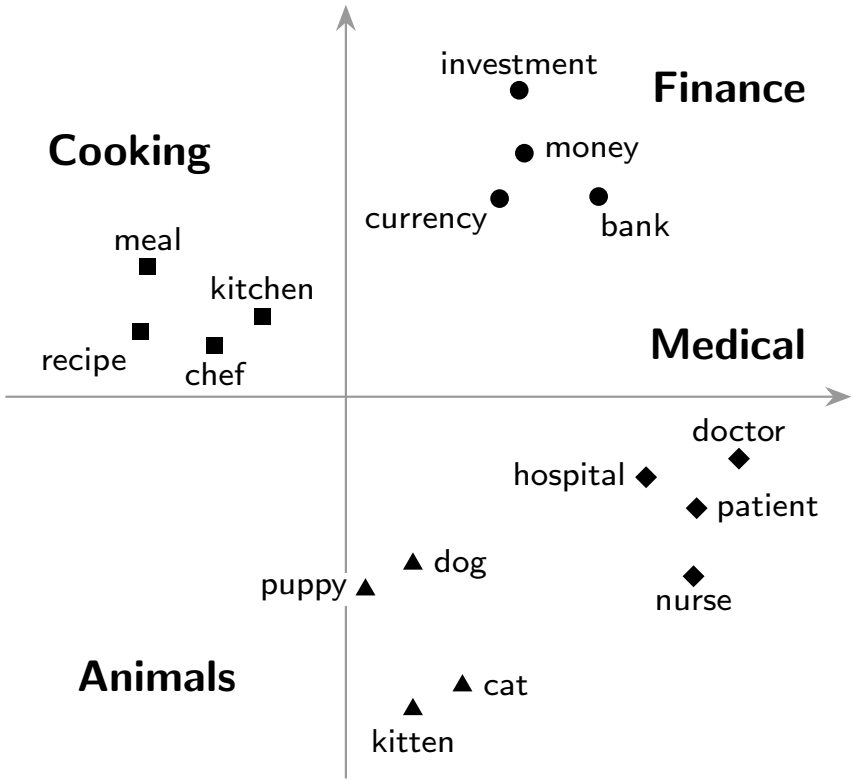
Columns are the same 16 words, in the same order.

How to read it

- Each cell is the **cosine similarity** of two words' embeddings; **darker** is more similar.
- The diagonal is 1: every word with itself.
- **Dark 4×4 blocks**: words on the same topic point nearly the same way, a **small angle**.
- **Light elsewhere**: words on different topics are **nearly perpendicular**.

The model was never told the topics. The blocks come only from how the words are used in text.

Similar Meanings Cluster Together



The same 16 embeddings, flattened to 2-D.

What the picture shows

- Each word's hundreds of numbers are **flattened to 2** so we can draw them, using **PCA** (the end of this unit) plus a method called t-SNE.
- Each topic lands in its own **cluster**: similar meaning, nearby points.
- The axes carry no units; only **who is near whom** matters.

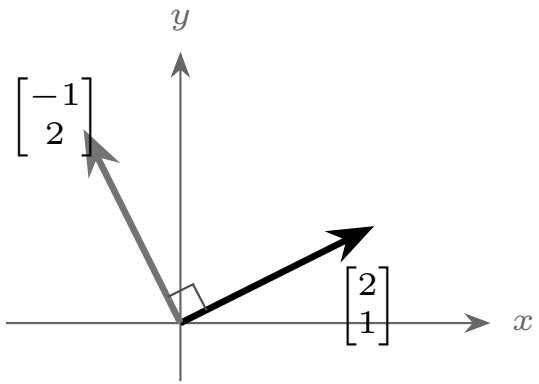
Back to the spam filter

Whole sentences get embeddings too. Now "win free cash now" and "claim complimentary money today" point nearly the **same way**, and the **same cosine** finds it. Search engines rank results this way.

Orthogonality

Definitions

- **Orthogonal** (perpendicular): $\bar{v} \cdot \bar{w} = 0$.
- **Orthonormal**: orthogonal **and** both unit vectors, $\|\bar{v}\| = \|\bar{w}\| = 1$.



- **Why**: $\cos 90^\circ = 0$.
- **Check**: $\begin{bmatrix}2\\1\end{bmatrix} \cdot \begin{bmatrix}-1\\2\end{bmatrix} = -2 + 2 = 0$.
- **Not** orthonormal: each has length $\sqrt{5}$. Normalize both and they are, like $\hat{\imath}$, $\hat{\jmath}$.
- No roots or trig needed: cheap even in $\mathbb{R}^{500}$.

Running examples

- **Velocity**: a pure **crosswind** adds nothing along the plane's heading; it only pushes the plane sideways.
- **Email**: two emails with **no words in common** are orthogonal.

Exercise: Dot Products and Angles

Try it

Let $\bar{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\bar{b} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, $\bar{c} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$.

1. Compute $\bar{a} \cdot \bar{b}$ and the angle between $\bar{a}$ and $\bar{b}$.
2. Compute $\bar{a} \cdot \bar{c}$. What does the answer say about them?
3. Compute $\bar{b} \cdot \bar{c}$. What does the **sign** say?
4. Turn $\bar{b}$ into a unit vector.

Exercise: Solution

Answers

1. $\bar{a} \cdot \bar{b} = (3)(3) + (0)(3) = 9$. With $\|\bar{a}\| = 3$ and $\|\bar{b}\| = \sqrt{18} = 3\sqrt{2}$, $\cos\theta = \frac{9}{3 \cdot 3\sqrt{2}} = \frac{1}{\sqrt{2}}$, so $\theta = 45°$. ($\bar{a}$ lies along the x-axis and $\bar{b}$ on the diagonal, so this is the answer the picture predicts.)
2. $\bar{a} \cdot \bar{c} = (3)(0) + (0)(-2) = 0$: they are **orthogonal**. One runs along the x-axis, the other along the y-axis.
3. $\bar{b} \cdot \bar{c} = (3)(0) + (3)(-2) = -6$. **Negative**, so the two point more than $90°$ apart: $\bar{b}$ heads up and right while $\bar{c}$ heads straight down.
4. $\hat{b} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$, and $0.5 + 0.5 = 1$ as required.

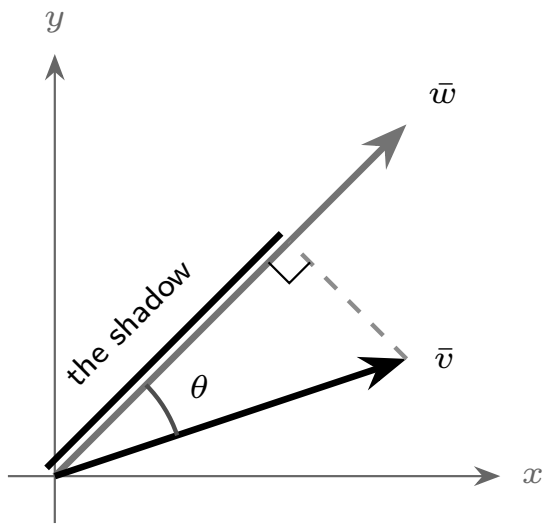
Projection: the Shadow of One Vector on Another

The **shadow** of $\bar{v}$ on the line through $\bar{w}$: the part of $\bar{v}$ running **along** $\bar{w}$.

Velocity: wind $\bar{v}$ on a plane heading northeast along $\bar{w}$, **not** along an axis:

$$\bar{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \hat{w} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

($\hat{w}$ is the unit vector from the last exercise.)



1. Dot with $\hat{w}$: a **number**

$$\text{comp}_{\bar{w}}(\bar{v}) = \bar{v} \cdot \hat{w} = \|\bar{v}\| \cos\theta$$

The **component** of $\bar{v}$ along $\bar{w}$: the shadow's **length**. Here $\frac{3}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.83$: the **tailwind**.

2. Times $\hat{w}$: a **vector**

$$\text{proj}_{\bar{w}}(\bar{v}) = (\bar{v} \cdot \hat{w})\,\hat{w}$$

The **projection**: that length, pointed along $\bar{w}$. Here $2\sqrt{2}\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$: the roots cancel.

Only **part** of the wind: the rest, the dashed drop, is the **crosswind** (next slide).

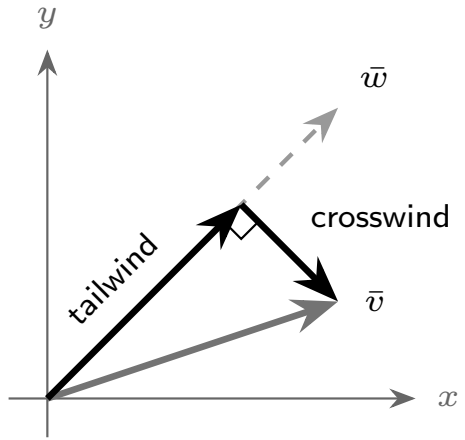
Notebook `U2-2_Vectors-3_ProjectionAndSimilarity`, §1, the drop drawn with its right angle

The Leftover: the Crosswind

3. Subtract: the **leftover**

$$\bar{v} - \text{proj}_{\bar{w}}(\bar{v}) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The part of the wind **across** the heading. It adds nothing to the speed along $\bar{w}$; it only shoves the plane **sideways**, off course.



True for **every** projection, not just this one

- **Perpendicular to** $\bar{w}$: $(1)(3) + (-1)(3) = 0$, a right angle.
- **The pieces rebuild** $\bar{v}$: $\begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, tailwind plus crosswind.
- **Pythagoras**: $\text{tail}^2 + \text{cross}^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 = 10 = \|\bar{v}\|^2$.

Look ahead: fitting a line to data (Unit 3) leaves errors that are this same kind of leftover, at a right angle to the fit.

Exercise: Projection and Similarity

Try it

1. Project $\bar{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ onto $\bar{w} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. Sketch it.
2. Two patients are recorded as $\bar{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\bar{q} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. What is their cosine similarity? Answer without computing anything.
3. Their Euclidean **distance**, on the other hand, is not small. Which of the two measures is the right one, and what does the choice depend on?

Exercise: Solution

Answers

1. $\|\bar{w}\| = 3$, so $\hat{w} = \frac{1}{3}\begin{bmatrix}3\\0\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\bar{v} \cdot \hat{w} = (2)(1) + (4)(0) = 2$, so $\text{proj}_{\bar{w}}(\bar{v}) = 2\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\0\end{bmatrix}$. Since $\bar{w}$ lies along the x-axis, the projection just keeps the x-component of $\bar{v}$ and drops the y-component, which is what the sketch shows.

2. **Exactly** 1**.** $\bar{q} = 3\bar{p}$, so the two vectors point along the same line in the same direction and the angle between them is 0. No arithmetic needed: scaling a vector never changes its direction, and cosine similarity sees direction only.

3. **It depends on whether size carries meaning.** If the components are counts (words in a document, visits in a year) then $\bar{q}$ is a bigger version of the same profile, and **cosine** is right. If they are clinical measurements, where a value three times higher is a genuinely different patient, then **distance** is right. Choosing between them is a modelling decision, not a mathematical one.

Linear Combinations, Span, and Independence

Linear Combinations

Definition

A **linear combination** of $\bar{v}_1, \ldots, \bar{v}_k$ scales each one and adds:

$$c_1\bar{v}_1 + c_2\bar{v}_2 + \cdots + c_k\bar{v}_k$$

The c_i are the **coefficients** or **weights**.

The two questions that matter

1. **Which points** can you reach?
2. For a given point, **which weights** get you there?

Running examples: we have been doing this all along

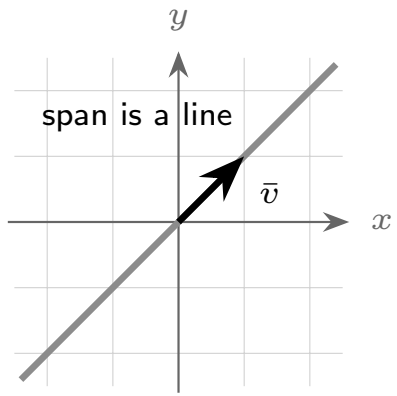
- **Velocity**: $\bar{v} = 150\hat{\imath} + 200\hat{\jmath}$, weights 150 and 200.
- **Email**: the word counts are the weights on the word unit vectors.
- **Patient**: the **average** of $\bar{p}$ and $\bar{q}$ has weights $\frac{1}{2}$, $\frac{1}{2}$:

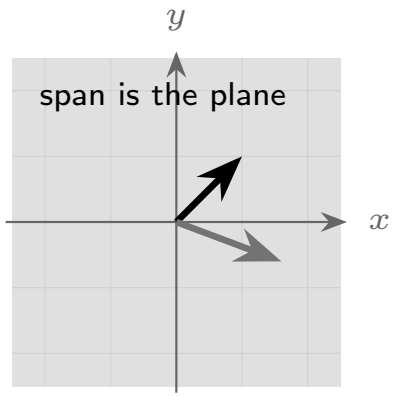
$$\frac{1}{2}\begin{bmatrix}54\\130\\6.1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}58\\138\\6.4\end{bmatrix} = \begin{bmatrix}56\\134\\6.25\end{bmatrix}$$

Span: Everything You Can Reach

Definition

The **span** of a set of vectors is the set of **all** their linear combinations.





In the plane there are three possibilities:

- **One nonzero vector**: a **line** through the origin.
- **Two, in different directions**: the **whole plane**.
- **Two on the same line**: only **that line**.

Every span contains $\bar{0}$: set all weights to zero.

Notebook `U2-2_Vectors-4_LinearCombinationsAndSpan`, §2, the span filled in point by point

Finding the Weights Is Solving a System

Is $\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ a combination of $\bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$? Write out what that would mean:

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \longleftrightarrow \quad \begin{bmatrix} a+b \\ b \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a+b=3 \\ b=2 \end{cases}$$

Solve it

- One **row** at a time: a Unit 2-1 system.
- Row 2: $b = 2$. Row 1: $a = 3 - 2 = 1$.
- So $\bar{u} = 1\bar{v} + 2\bar{w}$.

Check: $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

The three outcomes, as geometry

- **One solution**: the weights are **unique**.
- **No solution**: $\bar{u}$ lies **outside** the span.
- **Infinitely many**: the vectors are redundant, so many different linear combinations give the same $\bar{u}$.

Independence: Is Anything Redundant?

Definition: linearly dependent

A set of vectors is **linearly dependent** if one vector is a linear combination of the others. Two equivalent forms; **either works**:

$$\bar{u} = a\bar{v} + b\bar{w} \qquad \text{or} \qquad a\bar{u} + b\bar{v} + c\bar{w} = \bar{0} \text{ with } a, b, c \text{ not all } 0$$

- The first names the **redundant** vector, $\bar{u}$.
- The second treats all three alike (rearrange the first: weights $1, -a, -b$).
- "Not all 0" matters: all-zero weights give $\bar{0}$ for **any** vectors.

Definition: linearly independent

Not dependent: **no** vector is a combination of the others. Equivalently, $a\bar{u} + b\bar{v} + c\bar{w} = \bar{0}$ **only** when $a = b = c = 0$.

Worth memorizing: in $\mathbb{R}^n$, any **more than** n vectors are dependent.

Example: Testing for Dependence

Is $\bar{u}$ a combination of $\bar{v}$ and $\bar{w}$? Read the components one row at a time:

$$a\underbrace{\begin{bmatrix}1\\2\\0\end{bmatrix}}_{\bar{v}} + b\underbrace{\begin{bmatrix}0\\1\\1\end{bmatrix}}_{\bar{w}} = \underbrace{\begin{bmatrix}2\\3\\-1\end{bmatrix}}_{\bar{u}} \quad \longleftrightarrow \quad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = -1 \end{cases}$$

Solve it

- Row 1: $a = 2$.
- Row 2: $b = 3 - 2a = -1$.
- Row 3: $b = -1$. It **agrees**.

So $\bar{u} = 2\bar{v} - \bar{w}$, and the set is **dependent**: $\bar{u}$ adds nothing new.

Three equations, two unknowns, and they happened to be **consistent**. That is what "dependent" means.

Example: Change One Number and It Flips

Replace $\bar{u}$ by $\bar{u}' = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and ask the same question:

$$a \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \qquad \longleftrightarrow \qquad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = 0 \end{cases}$$

Solve it

- Rows 1 and 2 still force $a = 2$, $b = -1$.
- Row 3 demands $b = 0$: a **contradiction**.
- No weights exist. With $\bar{v}$, $\bar{w}$ not multiples of each other, the set is **independent**.

The takeaway

- Testing for dependence **is** solving a system.
- A solution is the redundancy; **no solution** means independent.
- $\bar{u}$ lay in the plane of $\bar{v}$ and $\bar{w}$; $\bar{u}'$ pokes out of it.

Why Dependence Is a Problem

A spare vector sounds harmless. It is not.

- **No unique answer.** If $\bar{u} = 2\bar{v} - \bar{w}$, every vector in the span of $\{\bar{u}, \bar{v}, \bar{w}\}$ is a linear combination of them in **infinitely many** ways.
- **Singular matrices.** Dependent columns make a matrix (next unit) with no inverse; `np.linalg.solve` refuses.
- **Nearly dependent is almost as bad.** Tiny changes in the data swing the weights wildly, and in floating point (Unit 1-5) you get huge coefficients, `inf`, or `NaN`.

Running example: collinear patient features

- Record each patient's height in inches **and** in centimetres: one column is $2.54\times$ the other.
- A regression cannot decide which one deserves the credit, so its coefficients go unstable.
- Fix: drop or combine the redundant column; PCA automates that.

Basis and Change of Basis

Basis and Dimension

Definition

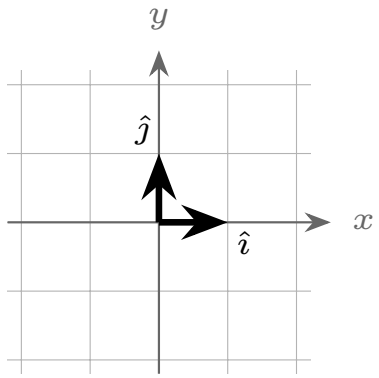
A **basis** is a set of vectors that is **independent** (nothing redundant) and **spanning** (reaches everything).

The **dimension** of the space is the number of vectors in a basis; every basis has the same number.

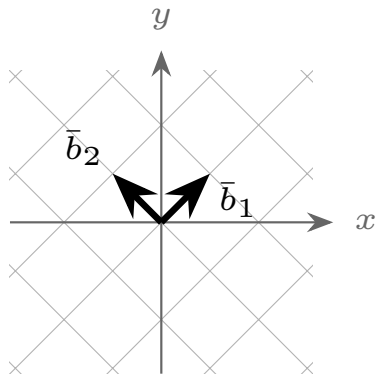
What to know

- **Minimal and complete**: drop one, you miss something; add one, it is redundant.
- **Not unique**: all four at right are bases. The first two are **orthonormal**, the nicest kind.

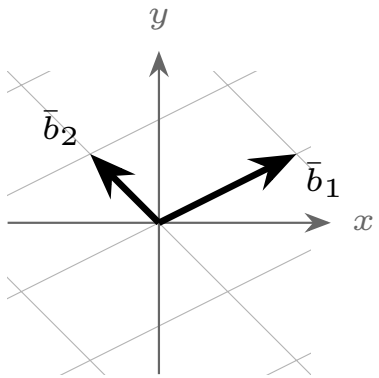
Four bases of $\mathbb{R}^2$, so $\mathbb{R}^2$ has dimension 2. Faint lines: the integer combinations $a\bar{b}_1 + b\bar{b}_2$, a grid over the **whole** plane.



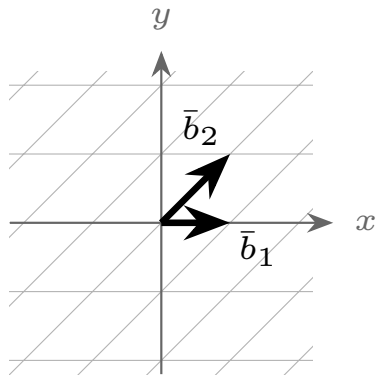
Standard: $\hat{\imath}$, $\hat{\jmath}$



Rotated $45°$



Skewed: $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$



Slanted: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Change of Basis: the Same Point, Different Coordinates

Coordinates are the **weights** on the basis vectors: $(2, 4)$ means $2\hat{\imath} + 4\hat{\jmath}$.

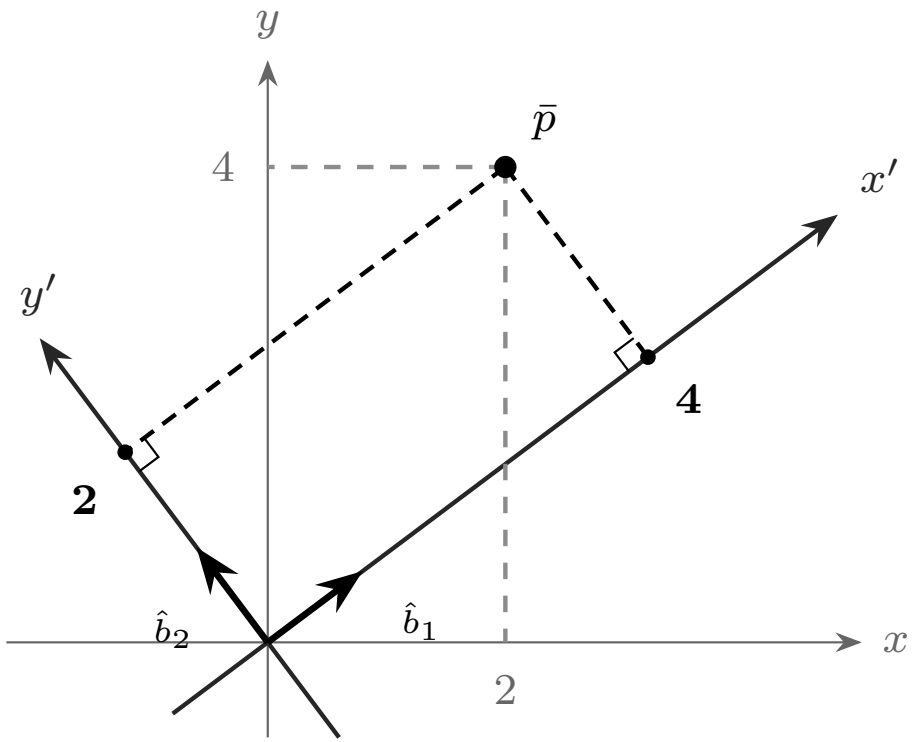
Rotated orthonormal axes

$$\hat{b}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, \quad \hat{b}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

New weights: projections

$$c_1 = \bar{p} \cdot \hat{b}_1 = (2)(0.8) + (4)(0.6) = 4$$
$$c_2 = \bar{p} \cdot \hat{b}_2 = (2)(-0.6) + (4)(0.8) = 2$$

Not orthonormal? Solve $c_1\bar{b}_1 + c_2\bar{b}_2 = \bar{p}$ instead.



Gray dashes: xy address $(2, 4)$
Black dashes: $x'y'$ address $(4, 2)$

Check: $4\hat{b}_1 + 2\hat{b}_2 = \begin{bmatrix} 3.2 \\ 2.4 \end{bmatrix} + \begin{bmatrix} -1.2 \\ 1.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. **Same point, new address.**

Notebook `U2-2_Vectors-5_IndependenceAndBasis`, §4, the same point read off two grids

Exercise: Span, Independence, and Basis

Try it

1. Is $\begin{bmatrix}5\\5\end{bmatrix}$ in the span of $\begin{bmatrix}1\\1\end{bmatrix}$? Is $\begin{bmatrix}1\\2\end{bmatrix}$?
2. Are $\begin{bmatrix}1\\3\end{bmatrix}$ and $\begin{bmatrix}2\\6\end{bmatrix}$ independent? What do they span?
3. Are $\begin{bmatrix}1\\1\end{bmatrix}$, $\begin{bmatrix}1\\-1\end{bmatrix}$, $\begin{bmatrix}2\\3\end{bmatrix}$ independent? Answer before computing.
4. Write $\begin{bmatrix}4\\0\end{bmatrix}$ in the basis $\bar{b}_1 = \begin{bmatrix}1\\1\end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix}1\\-1\end{bmatrix}$.

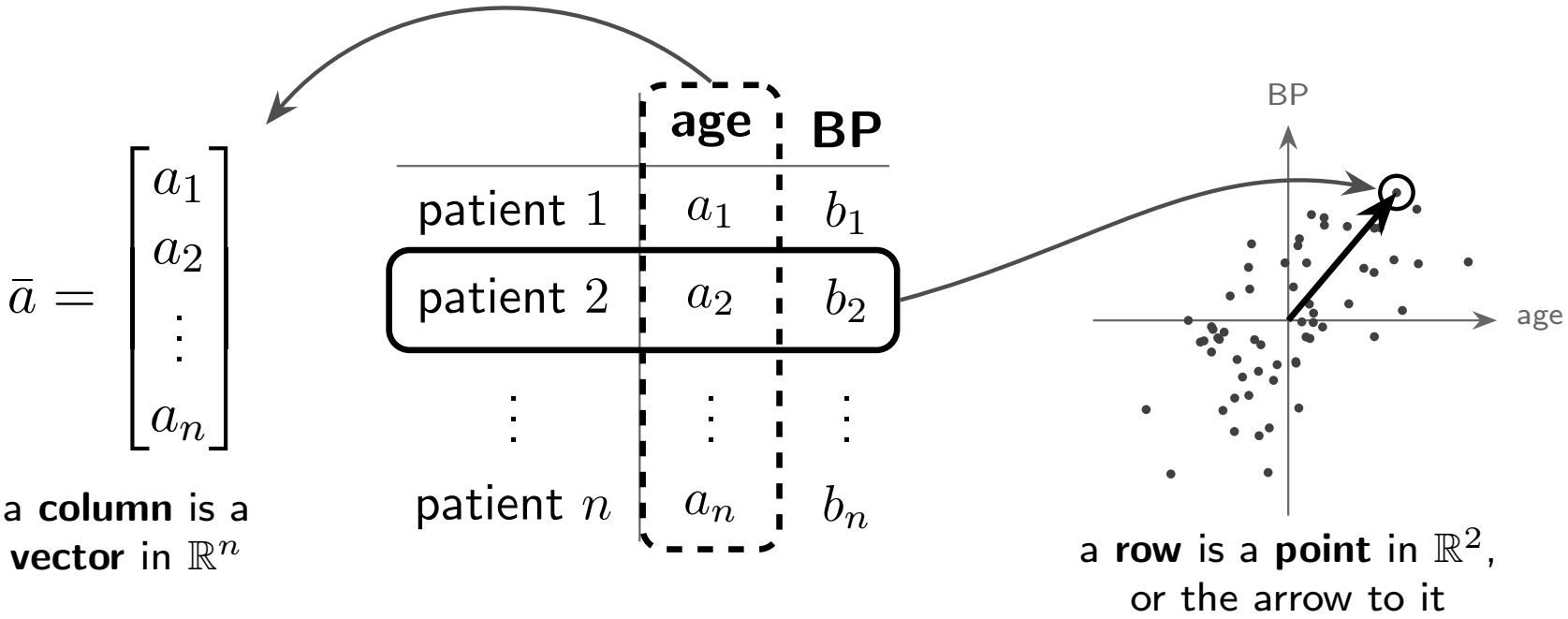
Exercise: Solution

Answers

1. **Yes** for the first: $\begin{bmatrix}5\\5\end{bmatrix} = 5\begin{bmatrix}1\\1\end{bmatrix}$. **No** for the second: the span of one vector is the line $y = x$, and $(1, 2)$ is not on it. There is no c with $c = 1$ and $c = 2$ at once.
2. **Dependent**, since the second is 2 times the first. They span only the line through the origin in the direction $\begin{bmatrix}1\\3\end{bmatrix}$, not the plane.
3. **Dependent**, and no computation is needed: three vectors in $\mathbb{R}^2$ are always dependent, because a basis for $\mathbb{R}^2$ has only two vectors. The first two already span the plane, so the third has to be a combination of them.
4. Solve $c_1\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}4\\0\end{bmatrix}$, that is $c_1 + c_2 = 4$ and $c_1 - c_2 = 0$. Adding gives $2c_1 = 4$, so $c_1 = c_2 = 2$: the coordinates are $(2, 2)$ in the $\bar{b}$ basis. (Note $\bar{b}_1 \cdot \bar{b}_2 = 0$, so this basis is orthogonal, just rotated from the standard one.)

Why Anyone Would Want a Different Basis

Running example: n patients, each with a standardized age a_i and blood pressure b_i. One table, two pictures:

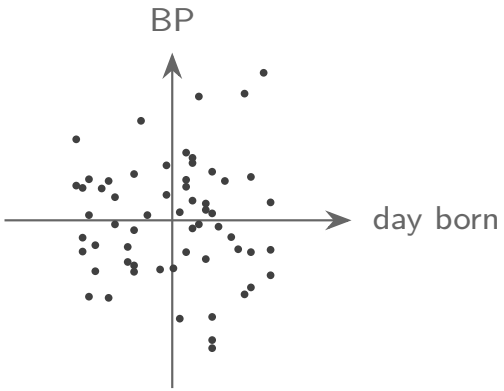


Dependence is about the columns

Is $\bar{b}$ a multiple of $\bar{a}$? No, so the columns are **independent**. But age and BP rise together, so in $\mathbb{R}^n$ the two columns **lean the same way**: the angle between them is well under $90°$.

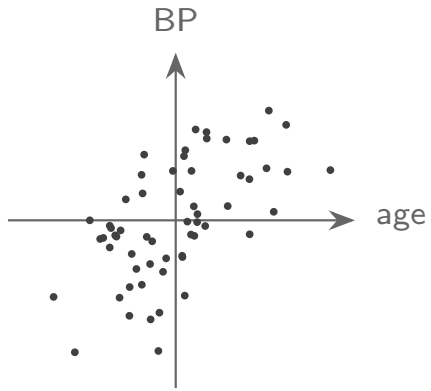
Three Tables, Three Clouds

Three tables like the last one, all standardized (so units drop out):



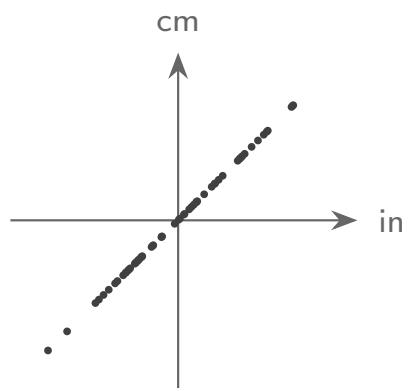
Unrelated

Day of the month born, and BP.
Columns: orthogonal.
Two patients: any angle; cosine similarity from -1 to 1.



Partly aligned

Age and BP.
Columns: independent, but lean the same way.
Two patients: often near parallel or opposite.



Dependent

Height in inches and in cm.
Columns: dependent (cm $= 2.54\times$ in).
Two patients: parallel or opposite; cosine similarity ± 1.

Rows and columns move together

Cosine similarity of two patients compares **rows**; dependence compares **columns**. The more dependent the columns, the closer every row lies to one line.

How Dependent? Pearson Correlation

Pearson correlation

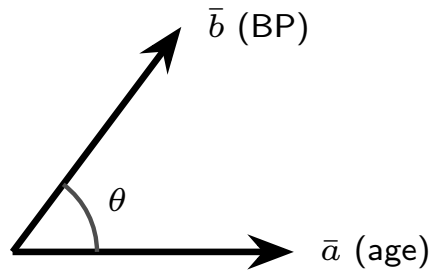
Let a_i, b_i be patient i's age and BP, **centered** (mean subtracted). Then

$$r = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2}\sqrt{\sum_{i=1}^n b_i^2}} = \frac{\bar{a} \cdot \bar{b}}{\|\bar{a}\| \, \|\bar{b}\|} = \cos\theta.$$

A sum over the **points** is a dot product of the **columns**: r is the cosine of the angle θ between $\bar{a}$ and $\bar{b}$ in $\mathbb{R}^n$.

Reading r

- $r = \pm 1$: columns **dependent**; points on a line.
- $r = 0$: columns **orthogonal**; no linear trend.
- The three clouds: about 0, about 0.6, $r = 1$.

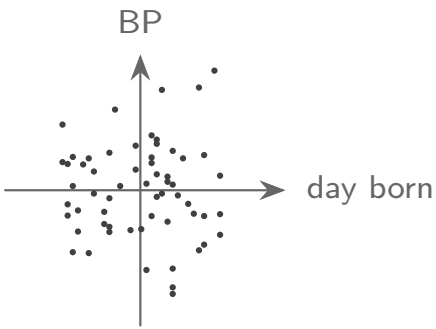


The age and BP columns, in the plane they span: $\cos\theta = r$.

r^2: Is a Feature Redundant?

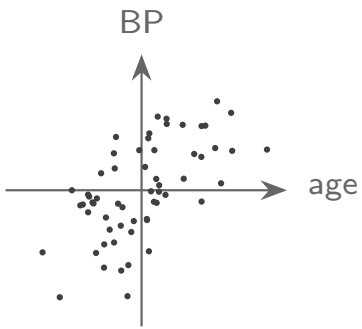
Square it

r^2 runs from 0 to 1 and ignores sign ($r = -1$ is as dependent as $r = 1$): the share of one column's spread that a straight line through the other explains.



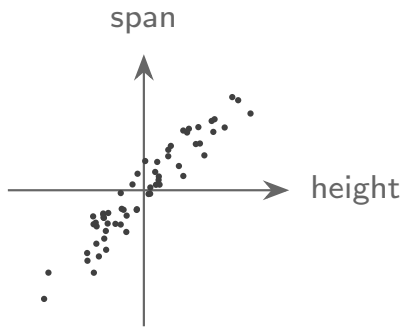
$r^2 \approx 0$

Day born and BP.
Unrelated: good.



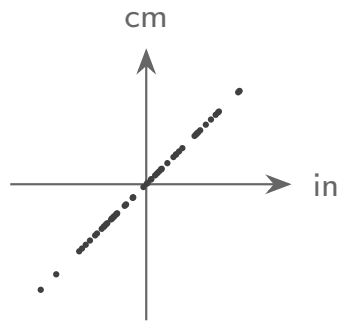
$r^2 \approx 0.36$

Age and BP.
Partly shared.



$r^2 \approx 0.90$

Height and arm span.
Nearly dependent:
still bad.



$r^2 = 1$

Height, in and cm.
Dependent: bad.

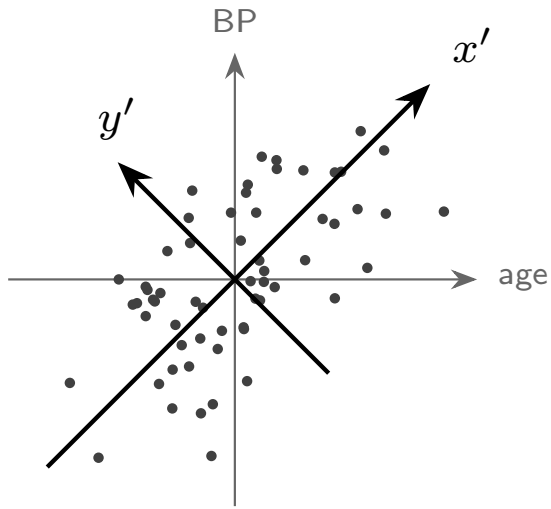
$r^2 = 0$: **good**, all new information ⟷ $r^2 = 1$: **bad**, redundant

Close to 1 is nearly as bad as 1: regression weights still swing wildly.
Careful: this is r^2 between two **features**. For a model's predictions against the truth, a **high** R^2 is good.

The Fix: Axes Along the Cloud

Change the basis

- Line the axes up with the cloud: x' carries most of the variation.
- y' carries much less, so **drop** it: two correlated columns become **one**, keeping most of the information.



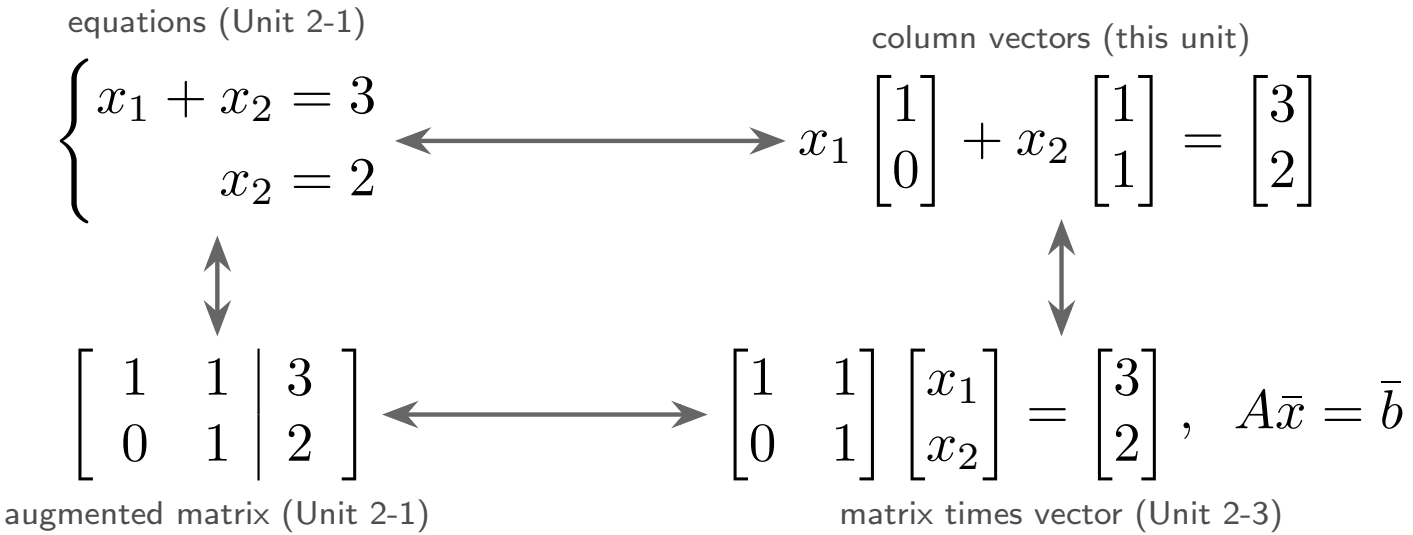
Age and BP, axes along the cloud.

That is PCA

- **Principal component analysis**, this unit's capstone, is a change of basis **chosen well**.
- Its new columns are **uncorrelated**, $r = 0$: the overlap between age and BP is gone.
- Which basis? **Eigenvectors** answer that, in §2.6.
- Same idea as compressing images and audio.

Looking Ahead: Where Matrices Come From

One System, Four Ways to Write It



Two ways to read it

- **Row picture**, read **across**: each equation is a line; the solution is where they cross.
- **Column picture**, read **down**: the unknowns weight the columns; the solution is the **weights** of the linear combination.

Where this is going: the columns side by side are a **matrix** A; Unit 2-3 defines $A\bar{x}$ as those columns weighted by $\bar{x}$.

Review

What we covered: vectors, operations, angles, and similarity

1. A **vector** has magnitude and direction. We draw it as an arrow and write it as a list of numbers, a **column** by default or a **row**; the two shapes differ, and Unit 2-3 says how.
2. A **point** is a location, a vector is a **displacement**; a vector may be slid anywhere without changing.
3. A vector with n components lives in $\mathbb{R}^n$, and **one row of a data table is one such vector**.
4. **Magnitude** $\|\bar{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$ is the Pythagorean hypotenuse.
5. **Addition** is head to tail, **scaling** stays on the vector's line, **subtraction** adds the reverse, and $\hat{v} = \bar{v}/\|\bar{v}\|$ is the **unit vector**.
6. **Distance** is $\|\bar{v} - \bar{w}\|$; it is scale-sensitive, so standardize first.
7. The **dot product** $\bar{v} \cdot \bar{w} = \sum v_i w_i$ multiplies matching components and adds; it returns a **number**, not a vector.
8. $\cos\theta = \dfrac{\bar{v} \cdot \bar{w}}{\|\bar{v}\|\|\bar{w}\|}$ gives the **angle**; **cosine similarity** ignores size.

Review

What we covered: projection, span, basis, and the look ahead

9. A dot product of 0 means **orthogonal**, and **projection** is the shadow of one vector on another.
10. A **linear combination** scales and adds; the **span** is everything reachable that way, and finding the weights is **solving a system** from Unit 2-1.
11. **Dependent** vectors are redundant (this is **collinearity**); independent ones each add a new direction.
12. A **basis** is independent and spanning, and its size is the **dimension**; **change of basis** re-describes the same point in different coordinates, which is the idea PCA is built on.
13. A system is also a **linear combination of column vectors**; collecting those columns into a grid gives a **matrix**, the subject of Unit 2-3.

Notebooks: `U2-2_Vectors-1_ArrowsAndOps`, `-2_DotProductAndAngle`, `-3_ProjectionAndSimilarity`, `-4_LinearCombinationsAndSpan`, `-5_IndependenceAndBasis`. **Previous:** Unit 2-1, Systems. **Next:** Unit 2-3, **Matrices**.